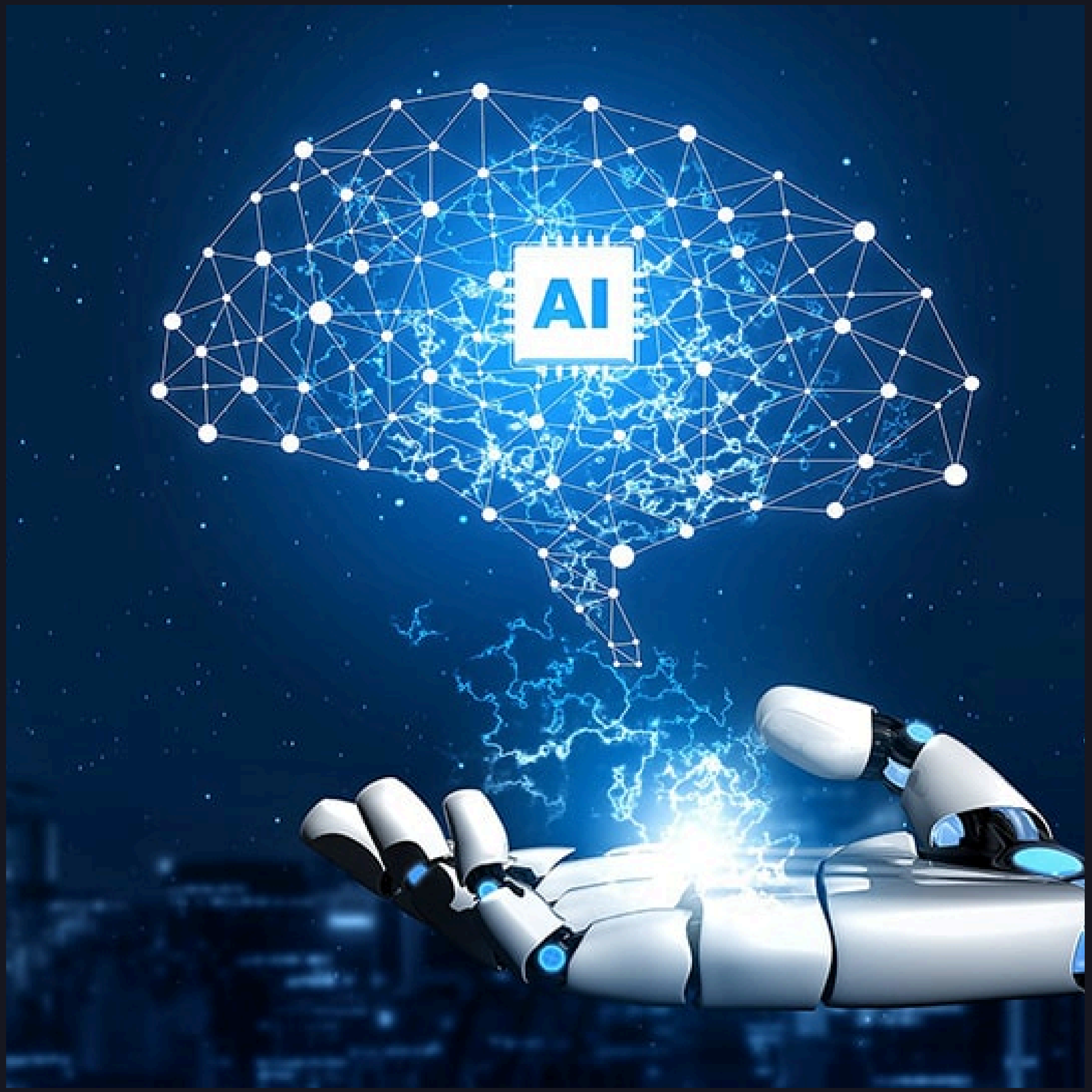
DATA ANALYTICS PROJECT

AI Adoption and Its Impact on Workforce Productivity

JUNY 2026

Created by Kreisna Witanto



ABOUT PROJECT

PROJECT INFORMATION

Duration : 5 Hours
Project Type : Exploratory Data Analysis (EDA) and Statistical Analysis
Dataset : AI Impact On Job Sector (Kaggle)

This project explores the impact of Artificial Intelligence (AI) adoption on workforce productivity using statistical analysis and exploratory data analysis (EDA). The analysis includes data preparation, descriptive statistics, correlation analysis, and data visualization to identify patterns and generate insights that support data-driven decision-making.

Business Understanding

Business Objectives

- Analyze the distribution of AI adoption across the workforce.
- Examine the relationship between AI adoption and employee productivity.
- Identify patterns and trends using descriptive statistics and exploratory data analysis (EDA).
- Generate data-driven insights to support decision-making regarding AI implementation.

Business Questions

- How is AI adoption distributed across different industries and job roles?
- Does AI adoption show a relationship with employee productivity?
- What patterns can be identified between AI usage and work-related performance metrics?

DATASET

Dataset Information

- Dataset: AI Impact On Job Sector
- Source: Kaggle
- Format: Csv
- Record: 2000 Records
- Columns: 20

SCREENSHOT DATASET

	A	B	C	D	E	F	G	H	I	J	K
1	Employee_ID,Age,Gender,Education_Level,Industry,Job_Role,Years_Experience,AI_Adoption_Level,Automation_Risk,										
2	E0001,50,Female,Bachelor,Marketing,Content Creator,26,High,High,Yes,106820,95455,Replaced,45,No,5,-10.64										
3	E0002,45,Male,High School,Manufacturing,Quality Inspector,19,Low,Low,Yes,74131,72013,Unchanged,36,Yes,6,19.05										
4	E0003,51,Female,Master,IT,DevOps Engineer,28,Medium,Medium,Yes,35311,42290,Modified,46,Yes,3,17.05										
5	E0004,48,Male,PhD,Education,Teacher,24,Medium,Medium,Yes,114478,107820,Modified,50,No,9,-2.47										
6	E0005,24,Male,Bachelor,Healthcare,Doctor,0,High,Medium,No,33890,40945,Modified,52,Yes,6,7.03										
7	E0006,46,Male,High School,Healthcare,Technician,21,Low,Medium,No,103969,104781,Modified,36,No,6,-19.04										
8	E0007,49,Male,PhD,Education,Administrator,24,High,Medium,No,78555,86183,Modified,48,Yes,3,-17.94										
9	E0008,57,Male,Bachelor,Finance,Auditor,34,Low,High,No,99479,105776,Unchanged,44,Yes,6,5.51										
10	E0009,50,Other,Bachelor,Education,Professor,27,Low,High,Yes,102409,104197,Unchanged,38,No,8,3.71										
11	E0010,52,Male,High School,Healthcare,Nurse,27,High,Low,Yes,31016,42699,Unchanged,47,Yes,4,30.67										
12	E0011,45,Female,High School,Retail,Store Manager,22,Medium,High,No,93704,92231,Unchanged,43,No,7,12.56										
13	E0012,36,Other,Master,Healthcare,Nurse,10,High,Low,Yes,78984,109956,Unchanged,46,No,9,26.33										
14	E0013,32,Other,Bachelor,Healthcare,Nurse,8,High,High,No,69504,85267,Modified,39,No,4,27.41										
15	E0014,49,Female,High School,Marketing,Content Creator,27,High,Low,Yes,60535,64752,Unchanged,39,Yes,5,-14.28										
16	E0015,43,Other,Master,Education,Professor,19,Medium,High,Yes,100592,102088,Unchanged,37,No,3,38.3										
17	E0016,26,Other,Bachelor,Education,Administrator,3,Low,Medium,Yes,36910,36482,Modified,49,Yes,9,-17.57										
18	E0017,47,Other,PhD,Education,Teacher,24,High,Medium,Yes,53419,46193,Modified,41,Yes,3,-18.11										
19	E0018,53,Other,Bachelor,Education,Teacher,28,High,High,Yes,48141,42537,Replaced,49,No,5,16.27										
20	E0019,50,Male,Bachelor,Education,Administrator,25,Medium,High,No,85820,79219,Unchanged,36,Yes,4,-2.61										
21	E0020,27,Female,PhD,Healthcare,Nurse,2,Medium,Low,No,99163,101637,Unchanged,41,No,8,33.0										

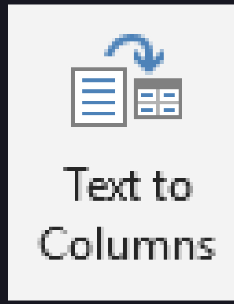
Variable	Format	Description
Employee_ID	General	Unique employee identifier.
Age	Number	Employee age in years.
Gender	Text	Employee gender.
Education_Level	Text	Highest education level attained.
Industry	Text	Employee's industry sector.
Job_Role	Text	Employee's job position.
Years_Experience	Number	Total years of work experience.
Experience_Group	Text	Experience level category.
AI_Adoption_Level	Text	Level of AI usage at work.
Automation_Risk	Text	Risk of job automation.
Upskilling_Required	Text	Whether upskilling is required.
Salary_Before_AI	Number	Salary before AI adoption.
Salary_After_AI	Number	Salary after AI adoption.
Salary_Change_%	Number (%)	Percentage change in salary.
Job_Status	Text	Current employment status.
Work_Hours_Per_Week	Number	Weekly working hours.
Remote_Work	Text	Remote work status.
Job_Satisfaction	Number	Employee satisfaction level.
Productivity_Change_%	Number (%)	Percentage change in productivity.
Work_Efficiency_Score	Number	Overall work efficiency score.

Microsoft Excel

DATA PREPARATION

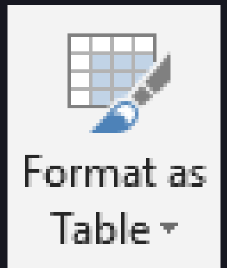
- Step 1: Text to Column
- Step 2: Format as Table
- Step 3: Standarized text values
- Step 4: Validate Numerical Values
- Step 5: Verify Calculated Values
- Step 6: Vaildate Experienced Group
- Step 7: Check Missing Values

Text to Column

Splitting or separating text data combined in a single column (cell) into multiple separate columns using : 

in excel.

Format as Table

Converted the raw CSV dataset into an Excel Table to simplify filtering, sorting, and structured data management using: 

or CTRL + T in excel.

Standarized Text Values

Standardized categorical values by removing extra spaces and applying consistent capitalization across text fields such as Gender, Industry, Job Role, and Education Level using formula =PROPER(TRIM(text)) in excel.

Validate Numerical Values

Validated numerical columns to identify missing values, non-numeric entries, and values outside logical ranges such as Age, Years_Experience, Salary_Before_AI, Salary_After_AI, Work_Hours_Per_Week, Work_Efficiency_Score before analysis using formula =ISNUMBER(value)

Verify Calculated Values

Recalculated Salary Change (%) to verify consistency between salary before and after AI adoption using formula:
=((Salary_After_AI-Salary_Before_AI)/Salary_Before_AI)*100.

Validate Experienced Group

Verified whether Experience_Group matched the corresponding Years_Experience category to ensure consistency across related variables. using formula =IF(G2<5,"Junior", IF(G2<=10,"Mid-Level", IF(G2<=20,"Senior","Expert"))).

Check Missing Values

Reviewed each variable to identify blank cells and ensure data completeness before statistical analysis. using formula:
=COUNTBLANK(range)

DATA QUALITY ASSESMENT

Overview

After completing the data cleaning process, a data quality assessment was conducted to verify that the dataset was complete, consistent, and suitable for statistical analysis. The assessment focused on data completeness, uniqueness, validity, and consistency.

Assessment Summary

- Missing Value ✔️
- Duplicate Record ✔️
- Data Types ✔️
- Category Consistency ✔️
- Numerical Vaidation ✔️
- Logical Consistency ✔️

1	Employee_ID	Age	Gender	Education_Level	Industry	Job_Role	Years_Experience	Experience_Group	AI_Adoption_Level	Automation_Risk	Upskilling_Required	Salary_Before_AI	Salary_After_AI
2	E0001	50	Female	Bachelor	Marketing	Content Crea	26	Expert	High	High	Yes	106820	9545
3	E0002	45	Male	High School	Manufactur	Quality Insp	19	Senior	Low	Low	Yes	74131	7201
4	E0003	51	Female	Master	IT	DevOps Engi	28	Expert	Medium	Medium	Yes	35311	4229
5	E0004	48	Male	PhD	Education	Teacher	24	Expert	Medium	Medium	Yes	114478	10782
6	E0005	24	Male	Bachelor	Healthcare	Doctor	0	Junior	High	Medium	No	33890	4094
7	E0006	46	Male	High School	Healthcare	Technician	21	Expert	Low	Medium	No	103969	10478
8	E0007	49	Male	PhD	Education	Administratc	24	Expert	High	Medium	No	78555	8618
9	E0008	57	Male	Bachelor	Finance	Auditor	34	Expert	Low	High	No	99479	10577
10	E0009	50	Other	Bachelor	Education	Professor	27	Expert	Low	High	Yes	102409	10419
11	E0010	52	Male	High School	Healthcare	Nurse	27	Expert	High	Low	Yes	31016	4269
12	E0011	45	Female	High School	Retail	Store Manag	22	Expert	Medium	High	No	93704	9223
13	E0012	36	Other	Master	Healthcare	Nurse	10	Mid-level	High	Low	Yes	78984	10995
14	E0013	32	Other	Bachelor	Healthcare	Nurse	8	Mid-level	High	High	No	69504	8526
15	E0014	49	Female	High School	Marketing	Content Crea	27	Expert	High	Low	Yes	60535	6475
16	E0015	43	Other	Master	Education	Professor	19	Senior	Medium	High	Yes	100592	10208
17	E0016	26	Other	Bachelor	Education	Administratc	3	Junior	Low	Medium	Yes	36910	3648
18	E0017	47	Other	PhD	Education	Teacher	24	Expert	High	Medium	Yes	53419	4619
19	E0018	53	Other	Bachelor	Education	Teacher	28	Expert	High	High	Yes	48141	4253
20	E0019	50	Male	Bachelor	Education	Administratc	25	Expert	Medium	High	No	85820	7921
21	E0020	27	Female	PhD	Healthcare	Nurse	2	Junior	Medium	Low	No	99163	10163
22	E0021	54	Other	High School	Manufactur	Quality Insp	32	Expert	Low	Medium	Yes	96842	9820
23	E0022	27	Male	PhD	Healthcare	Technician	3	Junior	Low	Medium	No	33748	3556
24	E0023	53	Other	Master	Education	Teacher	31	Expert	Low	High	No	84384	8098
25	E0024	23	Female	PhD	Retail	Store Manag	0	Junior	High	High	No	53664	4769
26	E0025	23	Female	High School	Finance	Accountant	0	Junior	Medium	High	No	30854	3034

Python

EXPLORATORY DATA ANALYSIS
(EDA)

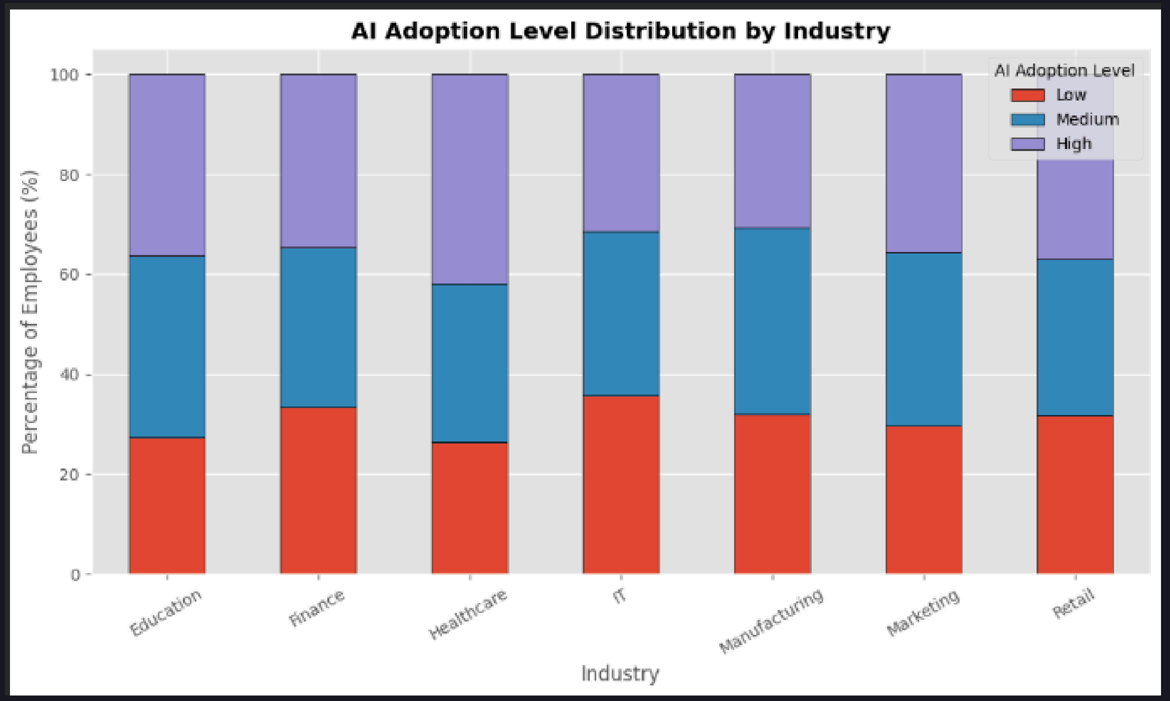
Business Questions

How is AI adoption distributed across different industries and job roles?

Objective

Analyze AI adoption across industries and job roles to identify sectors with the highest AI implementation.

AI Level Adoption by Industry



AI_Adoption_Level Industry	Low	Medium	High
Healthcare	26.3%	31.85%	41.85%
Retail	31.73%	31.37%	36.9%
Education	27.3%	36.52%	36.18%
Marketing	29.67%	34.8%	35.53%
Finance	33.33%	32.0%	34.67%
IT	35.57%	32.89%	31.54%
Manufacturing	31.86%	37.29%	30.85%

- Healthcare recorded the highest proportion of High AI Adoption (41.85%).
- IT had the highest percentage of employees in the Low AI Adoption (35.57%) category.
- Manufacturing was dominated by Medium AI Adoption (37.29%), suggesting moderate AI integration.
- Overall, the distribution of AI adoption is relatively balanced across industries, with no industry exceeding 50% in any adoption category.

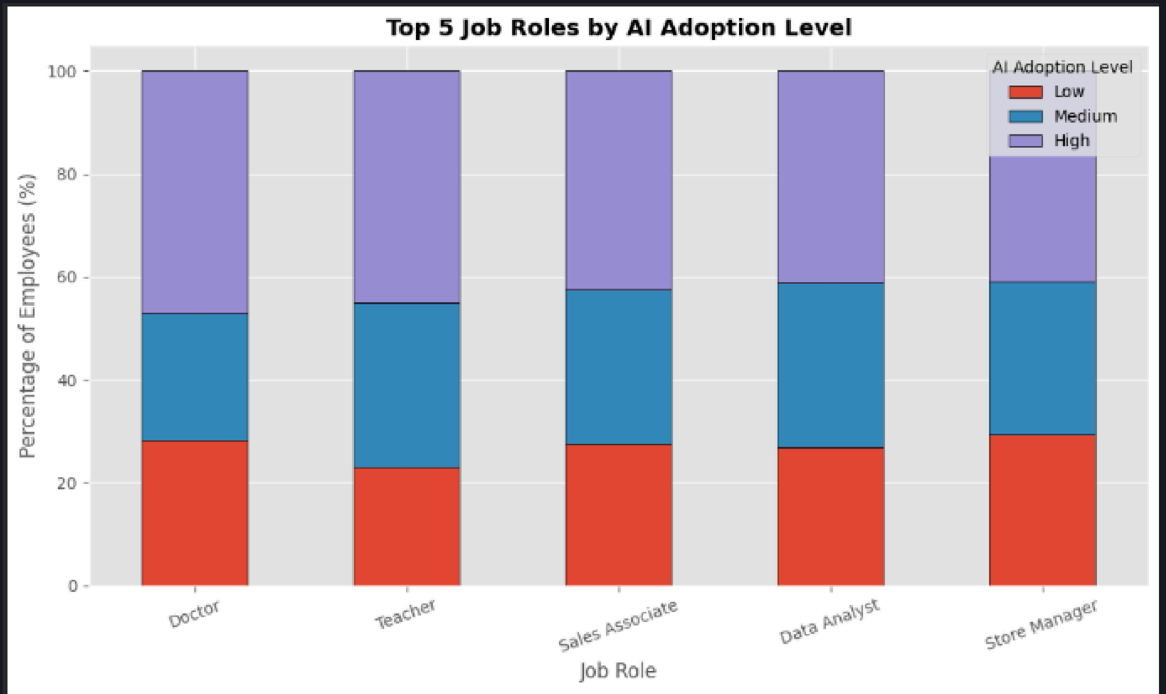
Business Questions

How is AI adoption distributed across different industries and job roles?

Objective

Analyze AI adoption across industries and job roles to identify sectors with the highest AI implementation.

AI Level Adoption by Top 5 Job Roles



AI_Adoption_Level	High
Job_Role	
Doctor	47.06%
Teacher	45.00%
Sales Associate	42.50%
Data Analyst	41.07%
Store Manager	40.95%

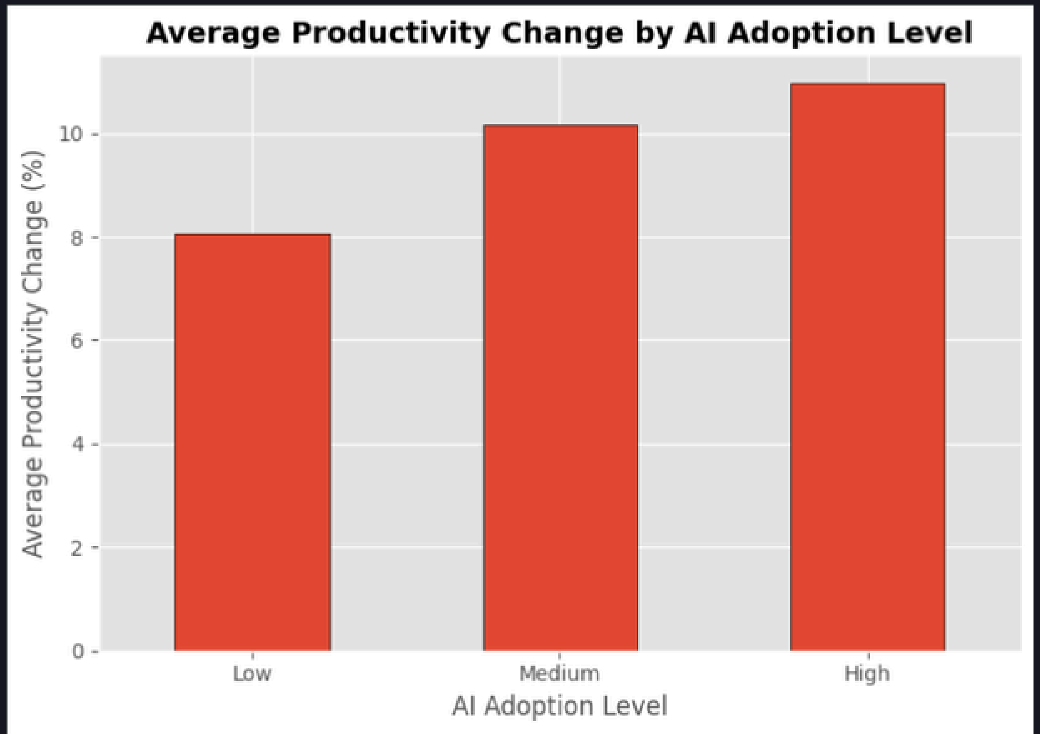
- Doctors recorded the highest proportion of High AI Adoption (47.06%).
- Teachers followed with 45.00%, while Sales Associates reached 42.50%.
- Auditors had the highest proportion of Low AI Adoption (42.42%), followed closely by Software Engineers (42.11%).
- Professors showed the highest proportion of Medium AI Adoption (42.86%).
- AI adoption differs considerably across occupations, indicating that implementation depends on job characteristics.

Business Questions

Does AI adoption show a relationship with employee productivity?

Objective

Analyze whether different levels of AI adoption are associated with changes in employee productivity




```
Reject H0
There is a statistically significant difference
```

```
F-statistic : 4.9780
P-value     : 0.0070
```

Test Method	F Statistic	P-Value	Decision	Conclusion
One Way ANOVA	4.978	0.007	Reject H0	Significant Difference Detected

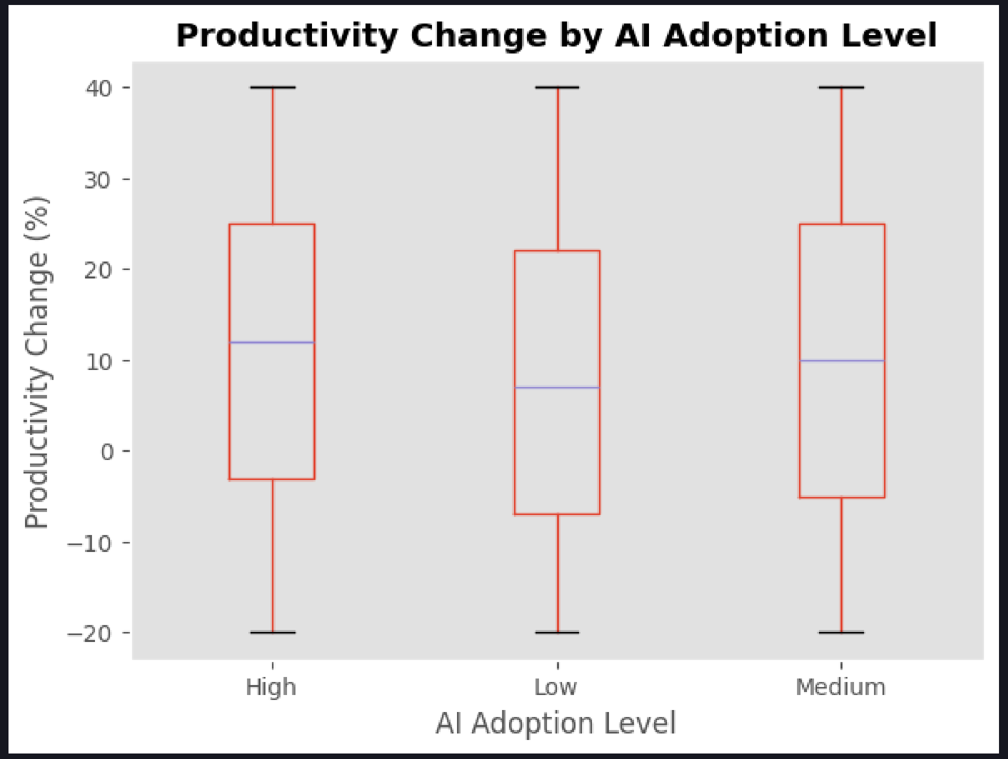
- High AI Adoption achieved the highest average productivity change (10.96%).
- ANOVA confirmed significant productivity differences across AI adoption levels (p = 0.007).
- Productivity differed significantly across AI adoption levels.
- Higher AI adoption was associated with greater productivity improvement.

Business Questions

Does AI adoption show a relationship with employee productivity?

Objective

Analyze whether different levels of AI adoption are associated with changes in employee productivity



- High AI Adoption shows a higher median productivity.
- Productivity varies within each adoption group.
- Although distributions overlap, ANOVA confirms the differences are statistically significant.

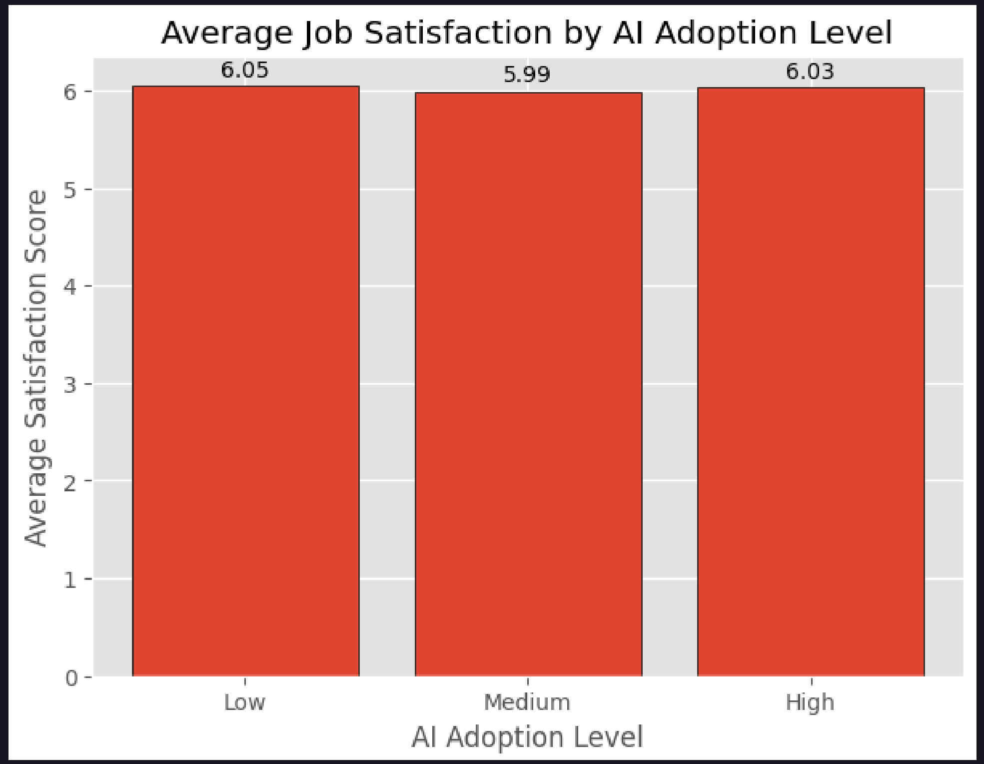
Business Questions

How does AI adoption influence employee job satisfaction and work efficiency?

Objective

Evaluate whether different levels of AI adoption are associated with employee job satisfaction and work efficiency.

Average Job Satisfaction by AI Adoption Level




```
F-statistic : 0.1571
P-value     : 0.8547
```

- Average job satisfaction remains relatively stable across all AI adoption levels (Low 6.05, Medium 5.99, High 6.03).
- No consistent upward or downward trend is observed between groups.
- ANOVA confirms this difference is not statistically significant (F=0.1571, p=0.8547).

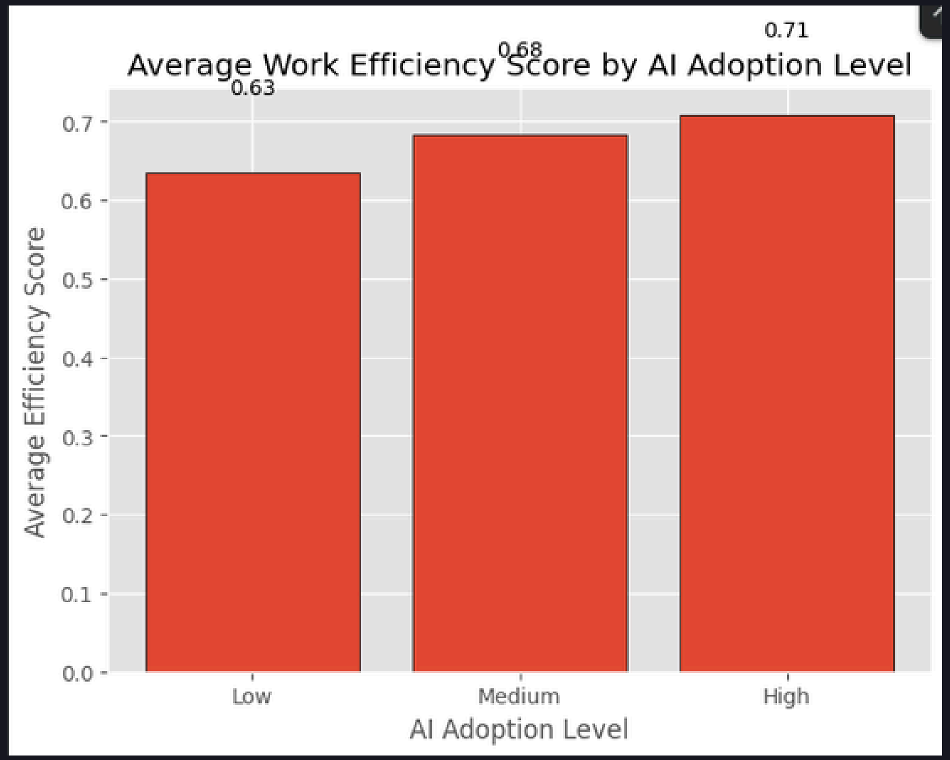
Business Questions

How does AI adoption influence employee job satisfaction and work efficiency?

Objective

Evaluate whether different levels of AI adoption are associated with employee job satisfaction and work efficiency.

Average Work Efficiency by AI Adoption Level




```
F-statistic : 5.5839
P-value     : 0.0038
```

- High AI Adoption shows the highest average work efficiency score (0.71), compared to Medium (0.68) and Low (0.63).
- Efficiency increases progressively with higher levels of AI adoption, suggesting a dose-response pattern.
- ANOVA confirms this difference is statistically significant (F=5.5839, p=0.0038).

Power BI

Data Visualization

Dashboard

Overview

Thank You

Drive Repository : https://bit.ly/4fjzWvC